

4.5 Exercises

- 4.5.1. a) The integers that are relatively prime to 60 serve as the generators for $\mathbb{Z}_{60}$. Since $60 = 2 \times 2 \times 3 \times 5$, we can omit all multiples of 2, 3, 5 as possible generators.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60

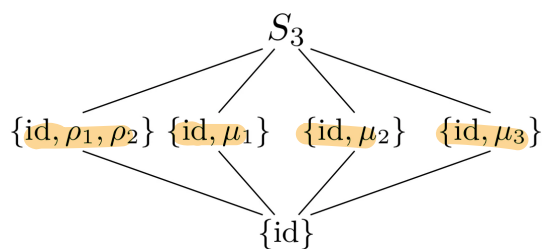
The remaining integers are relatively prime to 60, and are all prime.

- b) $U(8) = \{1, 3, 5, 7\}$. If $U(8)$ was cyclic, then there would exist some $x \in U(8)$ whose order was 4. However, $\text{order}(1) = 1$, and $\text{order}(3) = \text{order}(5) = \text{order}(7) = 2$. Hence $U(8)$ is not cyclic.

- c) $\mathbb{Q}$ is not cyclic, as there does not exist a single element $q := a/b$ such that $q = a/b$ generates all other rationals/fractions through repeated addition.

- d) CLAIM: If every proper subgroup of a group G is cyclic, then G is cyclic. This claim is **FALSE**... consider S_3 :

Each of the subgroups are cyclic, however, there is no single element that generates S_3 .



e) CLAIM: A group with a finite number of subgroups is finite.

We will use the following lemma:

Lemma: Every group G is a union of cyclic subgroups

Proof:

Observe that a group $G = \bigcup_{g \in G} \langle g \rangle$.

It follows then that we have the following chain of inclusions:

$$\{g\} \subset \langle g \rangle \subset G$$

$$\text{But then: } G = \bigcup_{g \in G} \langle g \rangle \subseteq \bigcup_{g \in G} \langle g \rangle \subseteq G$$

Hence: $G = \bigcup_{g \in G} \langle g \rangle$, since we have equality throughout.

Now suppose we have a finite number of subgroups, and consider an arbitrary element $g \in G$.

If the order of g is infinite in G , then the cyclic subgroup generated by g can establish a bijection with integers, $\mathbb{Z}$.

Note that $\mathbb{Z}$ has infinitely many subgroups.

This would imply that G also has infinitely many subgroups, but this contradicts our assumption.

If for all possible $g \in G$ their orders are finite, we now apply the lemma. The group G is a union of cyclic subgroups, and each of these are finite since element orders are finite.

Hence, G has finitely many subgroups $\Rightarrow G$ is a finite union of finite cyclic subgroups $\Rightarrow G$ must be finite.

4.5.2. a) The order of $5 \in \mathbb{Z}_{12} = 12$, since $5 \cdot 12 = 60 \equiv 0 \pmod{12}$

b) The order of $\sqrt{3} \in \mathbb{R} = \infty$; no integer k gives $(\sqrt{3})^k = 1$.

c) The order of $\sqrt{3} \in \mathbb{R}^* = \infty$; no integer k gives $(\sqrt{3})^k = 1$.

d) The order of $(-i) \in \mathbb{C}^* = 4$; $(-i)^4 = (i^2)(i^2) = (-1)(-1) = 1$

e) The order of $72 \in \mathbb{Z}_{240} = 10$;

$$240 = 2^4 \cdot 3 \cdot 5, \quad 72 = 2^3 \cdot 3^2 \Rightarrow \gcd(72, 240) = 2^3 \cdot 3 = 24$$

$$\text{Therefore, order}(72) = 240/24 = 10$$

f) The order of $312 \in \mathbb{Z}_{471} = 157$

$$471 = 3 \cdot 157, \quad 312 = 2^3 \cdot 3 \cdot 13 \Rightarrow \gcd(312, 471) = 3$$

$$\text{Therefore, order}(312) = 471/3 = 157.$$

4.5.3. a) $7\mathbb{Z} := \{\dots, -14, -7, 0, 7, 14, \dots\}$

b) $\langle 15 \rangle \subseteq \mathbb{Z}_{24} := \{0, 3, 6, 9, 12, 15, 18, 21\}$

c) All subgroups of $\mathbb{Z}_{12}$:

1. $\{0\}$

2. $\mathbb{Z}_{12} = \langle 1 \rangle = \langle 5 \rangle = \langle 7 \rangle = \langle 11 \rangle$

3. $\langle 2 \rangle := \{0, 2, 4, 6, 8, 10\}$

4. $\langle 3 \rangle := \{0, 3, 6, 9\}$

5. $\langle 4 \rangle := \{0, 4, 8\}$

6. $\langle 6 \rangle := \{0, 6\}$

d) All subgroups of $\mathbb{Z}_{60}$:

1. $\{0\}$

2. $\mathbb{Z}_{60} = \langle 1 \rangle = \langle i \rangle$, where $i \in \{7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 49, 53, 59\}$

3. $\langle 2 \rangle = \{0, 2, 4, \dots, 54, 56, 58\}$

4. $\langle 3 \rangle = \{0, 3, 6, \dots, 51, 54, 57\}$

5. $\langle 4 \rangle = \{0, 4, 8, \dots, 48, 52, 56\}$

6. $\langle 5 \rangle = \{0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55\}$

7. $\langle 6 \rangle = \{0, 6, 12, 18, 24, 30, 36, 42, 48, 54\}$

8. $\langle 10 \rangle = \{0, 10, 20, 30, 40, 50\}$

9. $\langle 12 \rangle = \{0, 12, 24, 36, 48\}$

10. $\langle 15 \rangle = \{0, 15, 30, 45\}$

11. $\langle 20 \rangle = \{0, 20, 40\}$

12. $\langle 30 \rangle = \{0, 30\}$

e) All subgroups of $\mathbb{Z}_{13}$: $\{0\}$ and $\mathbb{Z}_{13} = \langle 1 \rangle$

f) All subgroups of $\mathbb{Z}_{48}$:

1. $\{0\}$

2. $\mathbb{Z}_{48} = \langle 1 \rangle = \langle i \rangle$, where $i \in \{5, 7, 11, 13, 17, 19, 23, 25, 29, 31, 35, 37, 41, 43, 47\}$

3. $\langle 2 \rangle = \{0, 2, 4, \dots, 42, 44, 46\}$

4. $\langle 3 \rangle = \{0, 3, 6, \dots, 39, 42, 45\}$

5. $\langle 4 \rangle = \{0, 4, 8, \dots, 36, 40, 44\}$

6. $\langle 6 \rangle = \{0, 6, 12, 18, 24, 30, 36, 42\}$

7. $\langle 8 \rangle = \{0, 8, 16, 24, 32, 40\}$

8. $\langle 12 \rangle = \{0, 12, 24, 36\}$

9. $\langle 16 \rangle = \{0, 16, 32\}$

10. $\langle 24 \rangle = \{0, 24\}$

g) $\langle 3 \rangle \subseteq U(20) := \{3, 9, 7, 1\}$

* $3^3 = 27 \equiv 7 \pmod{20}$

* $3^4 = 21 \equiv 1 \pmod{20}$

h) $\langle 5 \rangle \subseteq U(18) := \{5, 7, 17, 13, 11, 1\}$

$$5^2 = 25 \equiv 7 \pmod{18}$$

$$5^3 = 5^2 \cdot 5 \equiv 17 \pmod{18}$$

$$5^4 = 5^3 \cdot 5 \equiv 13 \pmod{18}$$

$$5^5 = 5^4 \cdot 5 \equiv 11 \pmod{18}$$

$$5^6 = 5^5 \cdot 5 \equiv 1 \pmod{18}$$

i) Subgroup of $\mathbb{R}^*$ generated by $7 := \{7^k \mid k \in \mathbb{Z}\} = \{\dots, \frac{1}{49}, \frac{1}{7}, 1, 7, 49, \dots\}$

j) Subgroup of $\mathbb{C}^*$ generated by $i := \{i, -1, -i, 1\}$

k) Subgroup of $\mathbb{C}^*$ generated by $2i := \{(2i)^k \mid k \in \mathbb{Z}\} = \{\dots, \frac{1}{4}i, \frac{1}{2}, 1, 2i, -4, \dots\}$

l) Subgroup of $\mathbb{C}^*$ generated by $\frac{(1+i)}{\sqrt{2}} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$

Let $z := \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$. Note that z is a generator of the 8^{th} roots of unity.

Hence, $\langle z \rangle = \langle 1, z, z^2, z^3, z^4, z^5, z^6, z^7 \rangle$

m) Subgroup of $\mathbb{C}^*$ generated by $\frac{(1+\sqrt{3}i)}{2} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$

Let $z := \frac{1}{2} + \frac{\sqrt{3}}{2}i$. Note that z is a generator of the 6^{th} roots of unity.

Hence, $\langle z \rangle = \langle 1, z, z^2, z^3, z^4, z^5 \rangle$

- 4.5.4. a) Let $A := \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Then $\langle A \rangle = \{A, A^2 = I_2\} = \left\{ \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$
- b) Let $B := \begin{bmatrix} 0 & 1/3 \\ 3 & 0 \end{bmatrix}$. Then $\langle B \rangle = \{B, B^2 = I_2\} = \left\{ \begin{bmatrix} 0 & 1/3 \\ 3 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$
- c) Let $C := \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$. Then $\langle C \rangle = \{C, C^2, C^3, C^4, C^5, C^6 = I_2\}$
 $= \left\{ \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$
- d) Let $D := \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$. Then $\langle D \rangle = \{D, D^2, \dots, D^n, \dots\}$
 $= \left\{ \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}, \dots, \begin{bmatrix} 1 & -n \\ 0 & 1 \end{bmatrix}, \dots \right\}$
- e) Let $E := \begin{bmatrix} 1 & -1 \\ -1 & 0 \end{bmatrix}$. Then $\langle E \rangle = \{E, E^2, E^3, E^4, E^5, \dots\}$
 $= \left\{ \begin{bmatrix} 1 & -1 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}, \begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix}, \begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix}, \begin{bmatrix} 8 & -5 \\ -5 & 3 \end{bmatrix}, \dots \right\}$

* for any integer $n \geq 3$, we have $E_{ij}^n = E_{ij}^{n-1} + E_{ij}^{n-2}$, where $i, j \in \{1, 2\}$

f) Let $F := \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix}$. Then $\langle F \rangle = \{F, F^2, F^3, \dots, F^n, F^{12} = I_2\}$
 $= \left\{ \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix}, \begin{bmatrix} 1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & 1/2 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}, \begin{bmatrix} -\sqrt{3}/2 & 1/2 \\ -1/2 & -\sqrt{3}/2 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} -\sqrt{3}/2 & -1/2 \\ 1/2 & -\sqrt{3}/2 \end{bmatrix}, \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}, \begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

4.5.5. $\mathbb{Z}_{18} := \{0, 1, 2, \dots, 16, 17\}$

Possible element orders? Divisors of 18, i.e. 1, 2, 3, 6, 9, 18

Order 1 elements: $\{0\}$

Order 2 elements: $\{9\}$

Order 3 elements: $\{6, 12\}$

Order 6 elements: $\{3, 15\}$

Order 9 elements: $\{2, 4, 8, 10, 14, 16\}$

Order 18 elements: $\{1, 5, 7, 11, 13, 17\}$

4.5.6. Symmetries of a square:

- Identity : id
- 90° CW rotation : r_1
- 180° CW rotation : r_2
- 270° CW rotation : r_3
- Vertical axis reflection : v
- Horizontal axis reflection : h
- Diagonal reflection #1 (---): d_1
- Diagonal reflection #2 (---): d_2

$\circ$	id	r_1	r_2	r_3	v	h	d_1	d_2
id	id	r_1	r_2	r_3	v	h	d_1	d_2
r_1	r_1	r_2	r_3	id	d_1	d_2	h	v
r_2	r_2	r_3	id	r_1	h	v	d_2	d_1
r_3	r_3	id	r_1	r_2	d_2	d_1	v	h
v	v	d_2	h	d_1	id	r_2	r_3	r_1
h	h	d_1	v	d_2	r_2	id	r_1	r_3
d_1	d_1	v	d_2	h	r_1	r_3	id	r_2
d_2	d_2	h	d_1	v	r_3	r_1	r_2	id

ORDERS:

- $\text{order}(\text{id}) = 1$
- $\text{order}(x) = 2$, for $x \in \{v, h, d_1, d_2\}$
- $\text{order}(y) = 4$, for $y \in \{r_1, r_2, r_3\}$

4.5.7. $Q_8 := \{\pm 1, \pm I, \pm J, \pm K\}$

All cyclic subgroups of Q_8 :

1. $\langle 1 \rangle := \{1\}$
2. $\langle -1 \rangle := \{1, -1\}$

3. $\langle I \rangle := \{1, -1, I, -I\}$

$\hookrightarrow I, I^2 = -1, I^3 = -I, I^4 = 1$

4. $\langle J \rangle := \{1, -1, J, -J\}$

$\hookrightarrow J, J^2 = -1, J^3 = -J, J^4 = 1$

5. $\langle K \rangle := \{1, -1, K, -K\}$

$\hookrightarrow K, K^2 = -1, K^3 = -K, K^4 = 1$

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$J = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \quad K = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

4.5.8. $U(30) := \{1, 7, 11, 13, 17, 19, 23, 29\}$

All cyclic subgroups of $U(30)$:

1. $\langle 1 \rangle := \{1\}$

2. $\langle 11 \rangle := \{1, 11\}$

• $11^1 = 11$

• $11^2 = 121 \equiv 1 \pmod{30}$

4. $\langle 17 \rangle := \{1, 17, 19, 23\} = \langle 23 \rangle$

• $17^1 = 17$

• $17^2 = 289 \equiv 19 \pmod{30}$

• $17^3 = 17^2 \cdot 17 = 323 \equiv 23 \pmod{30}$

• $17^4 = 17^2 \cdot 17^2 = 361 \equiv 1 \pmod{30}$

3. $\langle 7 \rangle := \{1, 7, 13, 19\} = \langle 13 \rangle$

• $7^1 = 7$

• $7^2 = 49 \equiv 19 \pmod{30}$

• $7^3 = 7^2 \cdot 7 = 19 \cdot 7 = 133 \equiv 13 \pmod{30}$

• $7^4 = 7^3 \cdot 7 = 13 \cdot 7 = 91 \equiv 1 \pmod{30}$

5. $\langle 19 \rangle := \{1, 19\}$

• $19^1 = 19$

• $19^2 = 361 \equiv 1 \pmod{30}$

6. $\langle 29 \rangle := \{1, 29\}$

• $29^1 = 29$

• $29^2 = 841 \equiv 1 \pmod{30}$

4.5.9. $\mathbb{Z}_{32} := \{0, 1, 2, \dots, 30, 31\}$

We first require an element whose order is 8.

Note that the order $(4) = \frac{32}{\gcd(4, 32)} = \frac{32}{4} = 8$.

Therefore, $4 \in \mathbb{Z}_{32}$ generates a subgroup of order 8.

Specifically: $\langle 4 \rangle := \{0, 4, 8, 12, 16, 20, 24, 28\}$

For other generators "a", we still need $8 = \frac{32}{\gcd(a, 32)}$, i.e., $\gcd(a, 32) = 4$.

If $4|a$, then $a = 4k$ for $k \in \mathbb{Z}$.

Consider $a + 8$, so $a + 8 = 4k + 8$.

Suppose k is even, i.e. $k = 2m$. Then $4k + 8 = 8m + 8$, which implies divisibility by 8. But we want to ensure gcd is 4. Therefore, k must be odd.

Let $k := 2n + 1$.

Then $4k + 8 = 4(2n + 1) + 8 = 8n + 12 = 4(2n + 3)$

This verifies odd k preserves gcd divisibility by 4.

Hence, we obtain additional generators with $k = 3, 5, 7$

That is to say... $\langle 4 \rangle = \langle 4 \cdot 3 = 12 \rangle = \langle 4 \cdot 5 = 20 \rangle = \langle 4 \cdot 7 = 28 \rangle$

4.5.10. Finite order elements...

a) $\mathbb{Z}$? Just the additive identity: 0

c) $\mathbb{R}^*$? Just 1 and -1

b) $\mathbb{Q}^*$? Just 1 and -1

$$1 = \frac{1}{1}, \text{ so } 1 \cdot 1 = \left(\frac{1}{1}\right) \left(\frac{1}{1}\right) = 1$$

$$\text{Similarly, } -1 = \frac{1}{-1}, \text{ so } (-1)(-1) = \left(\frac{1}{-1}\right) \left(\frac{1}{-1}\right) = 1$$

4.5.11 If $a^{24} = e$ in a group G , the possible orders of $a \in G$ are:

- 1; in this case $a \in G$ would be the identity element: $(a')^{24} = e^{24} = e$
- 2; if $a^2 = e$, then $a^{24} = (a^2)^{12} = e^{12} = e$
- 3; if $a^3 = e$, then $a^{24} = (a^3)^8 = e^8 = e$
- 4; if $a^4 = e$, then $a^{24} = (a^4)^6 = e^6 = e$
- 6; if $a^6 = e$, then $a^{24} = (a^6)^4 = e^4 = e$
- 8; if $a^8 = e$, then $a^{24} = (a^8)^3 = e^3 = e$
- 12; if $a^{12} = e$, then $a^{24} = (a^{12})^2 = e^2 = e$
- 24; by definition of element order, and hypothesis.

4.5.12. Cyclic group with...

- 1 generator? $\mathbb{Z}_2$; generator is 1
- 2 generators? $\mathbb{Z}_3$; generators are 1, 2
- 4 generators? $\mathbb{Z}_5$; generators are 1, 2, 3, 4

If we want to find a cyclic group with n generators, we need to choose k such that for $\mathbb{Z}_k$, there are n integers relatively prime to k . That is to say: $\gcd(k, n) = 1$.

4.5.13.

n	$U(n)$	Cyclic?
1	$\{1\}$	YES: $\langle 1 \rangle = U(1)$
2	$\{1\}$	YES: $\langle 1 \rangle = U(2)$
3	$\{1, 2\}$	YES: $\langle 2 \rangle = U(3)$
4	$\{1, 3\}$	YES: $\langle 3 \rangle = U(4)$
5	$\{1, 2, 3, 4\}$	YES: $\langle 2 \rangle = U(5)$
6	$\{1, 5\}$	YES: $\langle 5 \rangle = U(6)$
7	$\{1, 2, 3, 4, 5, 6\}$	YES: $\langle 3 \rangle = U(7)$
8	$\{1, 3, 5, 7\}$	NO
9	$\{1, 2, 4, 5, 7, 8\}$	YES: $\langle 2 \rangle = U(9)$
10	$\{1, 3, 7, 9\}$	YES: $\langle 3 \rangle = U(10)$
11	$\{1, 2, \dots, 10\}$	YES: $\langle 2 \rangle = U(11)$
12	$\{1, 5, 7, 11\}$	NO
13	$\{1, 2, \dots, 11, 12\}$	YES: $\langle 2 \rangle = U(13)$
14	$\{1, 3, 5, 9, 11, 13\}$	YES: $\langle 3 \rangle = U(14)$
15	$\{1, 2, 4, 7, 8, 11, 13, 14\}$	NO
16	$\{1, 3, 5, 7, 9, 11, 13, 15\}$	NO
17	$\{1, 2, \dots, 15, 16\}$	YES: $\langle 3 \rangle = U(17)$
18	$\{1, 5, 7, 11, 13, 17\}$	YES: $\langle 5 \rangle = U(18)$
19	$\{1, 2, \dots, 17, 18\}$	YES: $\langle 2 \rangle = U(19)$
20	$\{1, 3, 7, 9, 11, 13, 17, 19\}$	NO

CLAIM: $U(n)$ is cyclic when...

- $n = 1$
- $n = p$, for prime $p \geq 2$
- $n = 2p$, for prime $p \geq 2$
- $n = p^k$, for prime $p \geq 3$
- $n = 2p^k$, for prime $p \geq 3$

Proof: TO DO

4.5.14. $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$: By 4.5.4 (a), the $\text{order}(A) = 2$.

$$B = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}; B^2 = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}; B^3 = B^2 B = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Therefore, the $\text{order}(B) = 3$.

$$AB = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$(AB)^2 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

$$(AB)^3 = (AB)^2 (AB) = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}$$

$\vdots$

$$(AB)^n = \begin{bmatrix} 1 & -n \\ 0 & 1 \end{bmatrix}; \text{hence, there does not exist } n \neq 0 \text{ such that } (AB)^n = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

4.5.15. a) $(3-2i) + (5i-6) = -3+3i$

d) $(9-i)(9-i) = (9-i)(9+i) = 81+9i-9i-i^2 = 82$

b) $(4-5i) - (4i-4) = (4-5i) - (4i+4) = 8-9i$

e) $i^{45} = i^{44} i = (i^4)^{11} \cdot i = (1)^{11} \cdot i = i$

c) $(5-4i)(7+2i) = 35+10i-28i-8i^2 = 43-18i$

f) $(1+i) + \overline{(1+i)} = (1+i) + (1-i) = 2$

4.5.16. a) $2\text{cis}(\pi/6) = 2[\cos(\pi/6) + i\sin(\pi/6)] = 2\left[\frac{\sqrt{3}}{2} + i\left(\frac{1}{2}\right)\right] = \sqrt{3} + i$

c) $3\text{cis}(\pi) = 3[\cos(\pi) + i\sin(\pi)] = -3$

b) $5\text{cis}(9\pi/4) = 5[\cos(9\pi/4) + i\sin(9\pi/4)] = 5[\cos(\pi/4) + i\sin(\pi/4)] = 5\left[\frac{\sqrt{2}}{2} + i\left(\frac{\sqrt{2}}{2}\right)\right] = \frac{5\sqrt{2}}{2} + \frac{5\sqrt{2}}{2}i$

d) $\frac{1}{2}\text{cis}(7\pi/4) = \frac{1}{2}[\cos(7\pi/4) + i\sin(7\pi/4)] = \frac{1}{2}\left[\frac{\sqrt{2}}{2} + i\left(-\frac{\sqrt{2}}{2}\right)\right] = \frac{\sqrt{2}}{4} - \frac{\sqrt{2}}{4}i$

$$* z = a + bi = r(\cos \theta + i \sin \theta) = r \operatorname{cis} \theta$$

$$+ \text{De Moivre: } (r \operatorname{cis} \theta)^n = r^n \cdot \operatorname{cis}(n\theta)$$

- 4.5.17. a) Cartesian: $1 - i$
Polar: $\sqrt{2} \operatorname{cis}(\frac{7\pi}{4})$
- b) Cartesian: $-5 = -5 + 0i$
Polar: $5 \operatorname{cis}(-\pi)$
- c) Cartesian: $2 + 2i$
Polar: $\sqrt{8} \operatorname{cis}(\frac{\pi}{4})$
- d) Cartesian: $\sqrt{3} + i$
Polar: $2 \operatorname{cis}(\frac{\pi}{6})$
- e) Cartesian: $-3i$
Polar: $3 \operatorname{cis}(\frac{3\pi}{2})$
- f) Cartesian: $2\sqrt{3} + 2i = 2(\sqrt{3} + i)$
Polar: $4 \operatorname{cis}(\frac{\pi}{6})$
- 4.5.18. a) For any nonzero complex $z = a + bi$, its multiplicative inverse is defined by: $z^{-1} = \frac{a - bi}{a^2 + b^2}$. Let $z = 1 + i$. Then $z^{-1} = (1 + i)^{-1} = \frac{1 - i}{1^2 + 1^2} = \frac{1 - i}{2} = \frac{1}{2} - \frac{1}{2}i$
- b) Note that $1 - i = z = \sqrt{2} \operatorname{cis}(\frac{7\pi}{4})$
By De Moivre's theorem, $(1 - i)^6 = z^6 = (\sqrt{2})^6 \cdot \operatorname{cis}(6 \cdot \frac{7\pi}{4})$

$$= 8 \operatorname{cis}(\frac{42\pi}{4})$$

$$= 8 \operatorname{cis}(\frac{21\pi}{2})$$

$$= 8(\cos(\frac{21\pi}{2}) + i \sin(\frac{21\pi}{2}))$$

$$= 8i$$
- c) $\sqrt{3} + i = z = 2 \operatorname{cis}(\frac{\pi}{6})$

$$(\sqrt{3} + i)^5 = z^5 = 2^5 \cdot \operatorname{cis}(\frac{5\pi}{6})$$

$$= 32(\cos(\frac{5\pi}{6}) + i \sin(\frac{5\pi}{6}))$$

$$= 32(-\frac{\sqrt{3}}{2} + \frac{1}{2}i)$$

$$= -16\sqrt{3} + 16i$$
- d) $(-i)^{10} = [(-i)^2]^5 = 1^5 = 1$
- e) $\left(\frac{1 - i}{2}\right)^4 = \frac{(1 - i)^4}{2^4} = \frac{(\sqrt{2})^4 \cdot \operatorname{cis}(4 \cdot \frac{7\pi}{4})}{16} = \frac{\operatorname{cis}(7\pi)}{4} = -\frac{1}{4}$
- f) $(-\sqrt{2} - \sqrt{2}i) = z = 2 \operatorname{cis}(\frac{5\pi}{4})$

$$(-\sqrt{2} - \sqrt{2}i)^{12} = z^{12} = 2^{12} \operatorname{cis}(12 \cdot \frac{5\pi}{4})$$

$$= 4096 \operatorname{cis}(15\pi)$$

$$= -4096$$
- g) $(-2 + 2i)^{-5} = [(-2 + 2i)^5]^{-1} = (128 - 128i)^{-1} = \frac{128 + 128i}{128^2 + 128^2} = \frac{128(1 + i)}{2 \cdot 128^2}$

$$= \frac{(1 + i)}{256}$$

$$= \frac{1}{256} + \frac{1}{256}i$$
- Let $z = -2 + 2i = \sqrt{8} \operatorname{cis}(\frac{3\pi}{4})$

$$\Rightarrow z^5 = (\sqrt{8})^5 \cdot \operatorname{cis}(\frac{15\pi}{4})$$

$$= 128\sqrt{2}(\cos(\frac{15\pi}{4}) + i \sin(\frac{15\pi}{4}))$$

$$= 128 - 128i$$

4.5.19. Let $z = a + bi$, so $\bar{z} = a - bi$

a) $|\bar{z}| = |a - bi| = |a| + |-bi| = |a| + |b| = |a + bi| = |z|$ $\square$

b) $|z|^2 = |z| \cdot |z|$
 $= (\sqrt{a^2 + b^2}) (\sqrt{a^2 + b^2})$
 $= a^2 + b^2$

$z\bar{z} = (a + bi)(a - bi)$
 $= a^2 - \cancel{abi} + \cancel{abi} - b^2 i^2$
 $= a^2 + b^2$

Therefore: $|z|^2 = a^2 + b^2 = z\bar{z}$ $\square$

c) By definition, for $z = a + bi$, we have that: $z^{-1} = \frac{a - bi}{a^2 + b^2}$.

But $a - bi = \bar{z}$, and $|z|^2 = a^2 + b^2$ by part (b).

Hence: $z^{-1} = \frac{\bar{z}}{|z|^2}$ $\square$

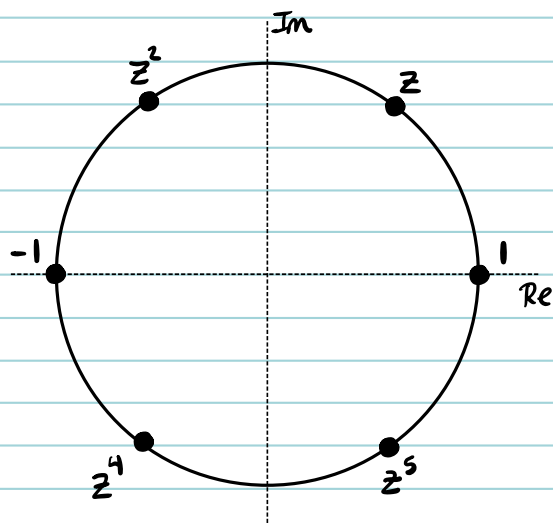
Let $w = c + di$, so $\bar{w} = c - di$

d) $|z + w| = |(a + bi) + (c + di)| \leq |a + bi| + |c + di| \leq |z| + |w|$ $\square$

e) $||z| - |w|| = ||a + bi| - |c + di|| = |(a + bi) - (c + di)| \leq |z - w|$ $\square$

f) $|zw| = |(a + bi)(c + di)| = |(a + bi)| \cdot |(c + di)| = |z||w|$ $\square$

4.5.20.



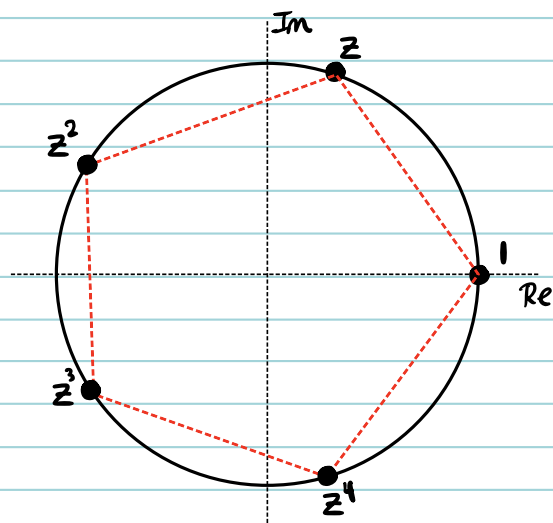
k	Root of unity
0	$z^0 = 1$
1	$z^1 = \text{cis}(2\pi/6) = (\frac{1}{2}) + i(\frac{\sqrt{3}}{2})$
2	$z^2 = \text{cis}(4\pi/6) = (-\frac{1}{2}) + i(\frac{\sqrt{3}}{2})$
3	$z^3 = -1$
4	$z^4 = \text{cis}(8\pi/6) = (-\frac{1}{2}) - i(\frac{\sqrt{3}}{2})$
5	$z^5 = \text{cis}(10\pi/6) = (\frac{1}{2}) - i(\frac{\sqrt{3}}{2})$

Generators are the primitive roots of unity.

The primitive roots are:

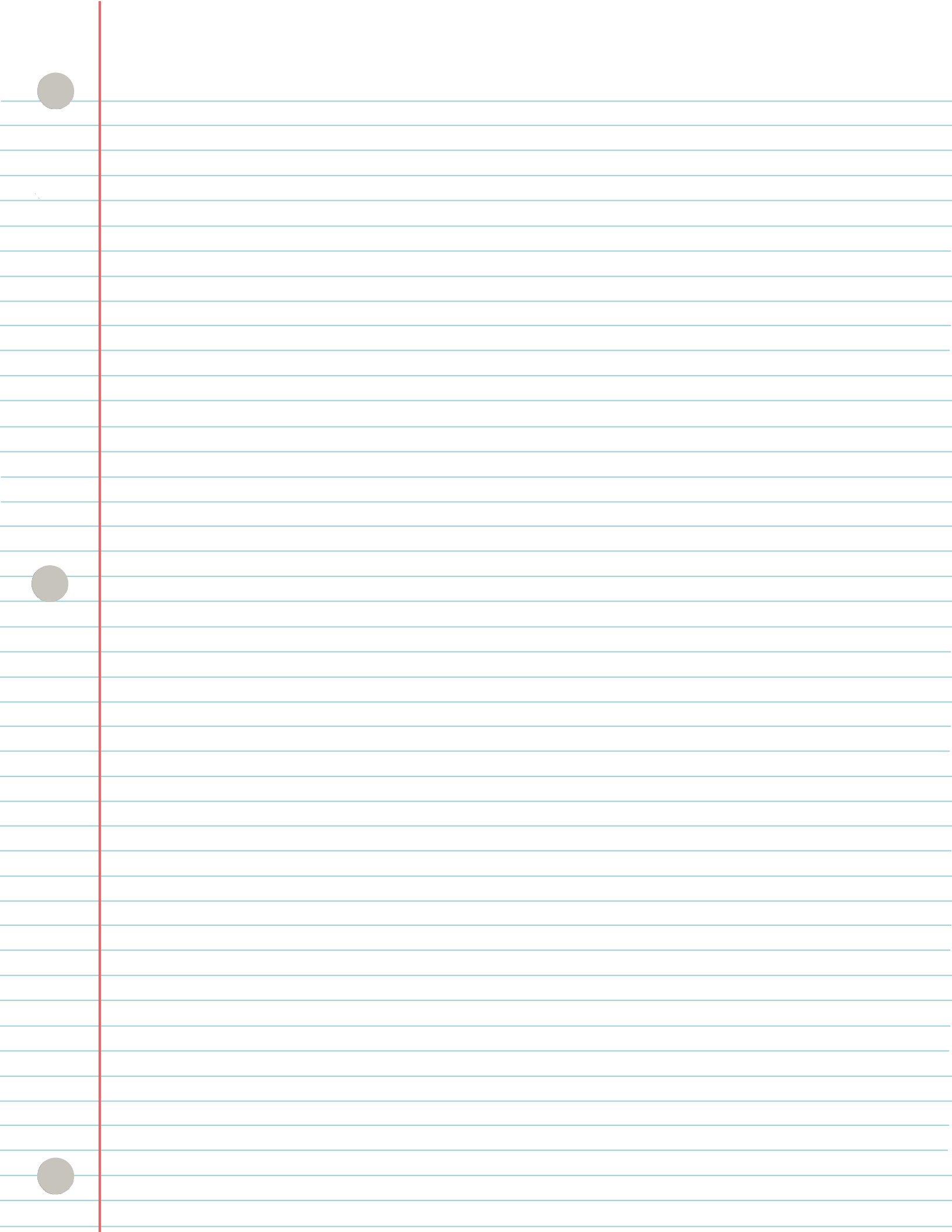
$\bullet z^1 = \text{cis}(\pi/3)$ $\bullet z^5 = \text{cis}(5\pi/3)$

4.5.21.



k	Root of unity
0	$z^0 = 1$
1	$z^1 = \text{cis}(2\pi/5) = e^{2\pi i/5}$
2	$z^2 = \text{cis}(4\pi/5) = e^{4\pi i/5}$
3	$z^3 = \text{cis}(6\pi/5) = e^{6\pi i/5}$
4	$z^4 = \text{cis}(8\pi/5) = e^{8\pi i/5}$

Note that $\text{gcd}(k, 5) = 1$ for any nonzero k less than n . Therefore, z^1, z^2, z^3 , and z^4 all generate the 5th roots of unity and are primitive.



4.5.21. List and graph the 5th roots of unity. What are the generators of this group? What are the primitive 5th roots of unity?

4.5.22. Calculate each of the following.

(a) $292^{3171} \pmod{582}$

(c) $2071^{9521} \pmod{4724}$

(b) $2557^{341} \pmod{5681}$

(d) $971^{321} \pmod{765}$

4.5.23. Let $a, b \in G$. Prove the following statements.

(a) The order of a is the same as the order of a^{-1} .

(b) For all $g \in G$, $|a| = |g^{-1}ag|$.

(c) The order of ab is the same as the order of ba .